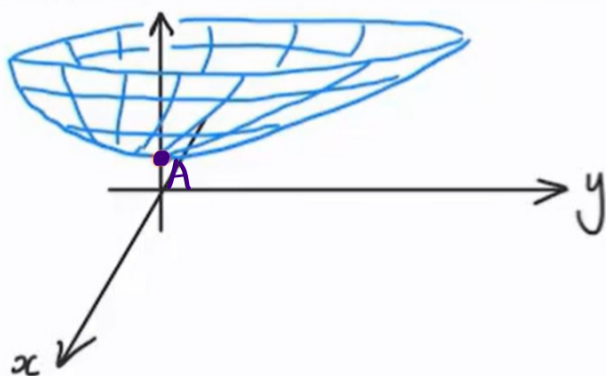


Critical Points



Point A is a critical point / stationary point / local minima. It has zero slope in every direction $\left[\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} = 0 \right]$
 $\Rightarrow \nabla f = 0$

Example 1

What are the critical points of $f(x,y) = -(x^2-1)(y^2-1)$?

$$\frac{\partial f}{\partial x} = -2x(y^2-1)$$

$$-2x(y^2-1) = 0$$

$$\therefore x = 0 \text{ or } y = \pm 1$$

$$\frac{\partial f}{\partial y} = -2y(x^2-1)$$

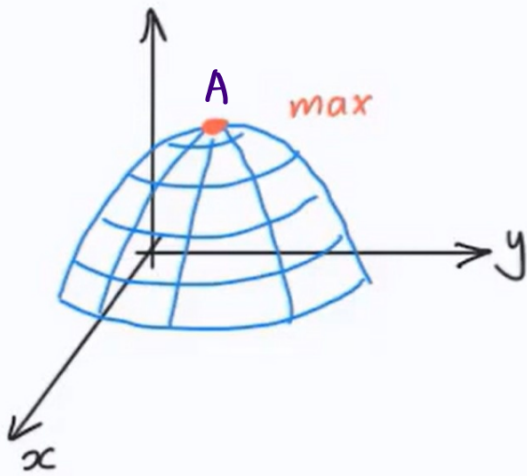
$$-2y(x^2-1) = 0$$

$$\therefore y = 0 \text{ or } x = \pm 1$$

$\therefore$ Solutions :

$$(0,0), (1,1), (1,-1), (-1,1), (-1,-1)$$

Characterising critical points

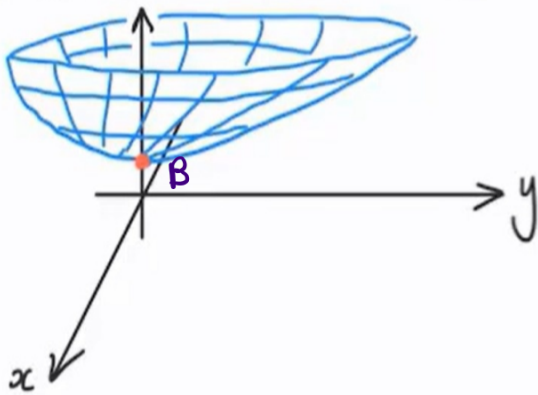


At point A, the change in slope
in any positive direction is negative

[from / to \]

$$\hookrightarrow \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial x} \right] < 0, \quad \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial y} \right] < 0$$

$$\therefore \frac{\partial^2 f}{\partial u^2} < 0$$



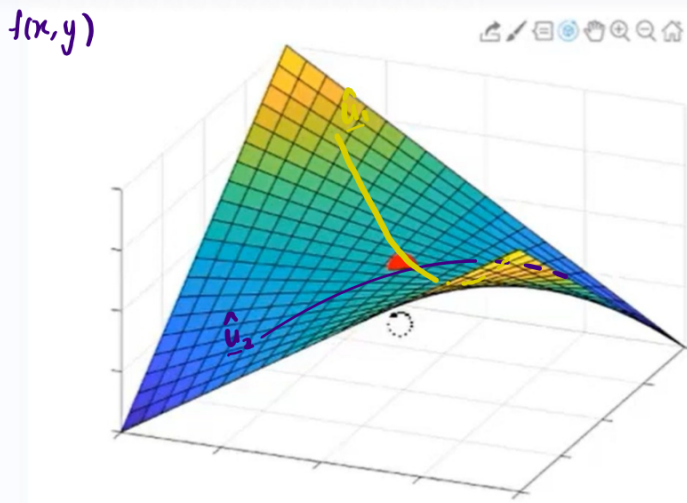
At point B, the change in slope
in any positive direction is positive

[from \ to /]

$$\hookrightarrow \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial x} \right] > 0, \quad \frac{\partial}{\partial y} \left[\frac{\partial f}{\partial y} \right] > 0$$

$$\therefore \frac{\partial^2 f}{\partial u^2} > 0$$

Saddle



For a saddle, there is at least one direction where the point is a max, and one direction where the point is a minimum.

[Example: $\frac{\partial f}{\partial u_1} < 0$, $\frac{\partial f}{\partial u_2} > 0$]

To determine whether the function ^{gradient} increases/decreases, we use the "1st order directional derivative", $\frac{\partial f}{\partial u}$. Similarly, to determine whether the function ^{curvature} $\frac{\partial f}{\partial u}$ increases/decreases, we use the "1nd order directional derivative", $\frac{\partial^2 f}{\partial u^2}$.

$$\therefore D_{uf}^1 = \frac{\partial}{\partial u} \left[\frac{\partial f}{\partial u} \right] \text{ Directional Derivative [ROC along } u \text{]}$$

$$= \frac{\partial}{\partial u} \left[\hat{u} \cdot \nabla f \right]$$

$$= \frac{\partial}{\partial u} \left[\begin{pmatrix} \hat{u}_1 \\ \hat{u}_2 \end{pmatrix} \cdot \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \end{pmatrix} \right]$$

$$= \frac{\partial}{\partial u} \left[\hat{u}_1 \frac{\partial f}{\partial x} + \hat{u}_2 \frac{\partial f}{\partial y} \right]$$

$$= \hat{u} \cdot \nabla \left[\hat{u}_1 \frac{\partial f}{\partial x} + \hat{u}_2 \frac{\partial f}{\partial y} \right] \text{ Curvature vector } \left[\frac{\partial}{\partial u} \square = \hat{u} \cdot \nabla \square \right]$$

$$= \begin{pmatrix} \hat{u}_1 \\ \hat{u}_2 \end{pmatrix} \cdot \begin{pmatrix} \hat{u}_1 \frac{\partial^2 f}{\partial x^2} + \hat{u}_2 \frac{\partial^2 f}{\partial x \partial y} \\ \hat{u}_1 \frac{\partial^2 f}{\partial y \partial x} + \hat{u}_2 \frac{\partial^2 f}{\partial y^2} \end{pmatrix} \text{ Converting to}$$

↓ matrix form

$$= \begin{pmatrix} \hat{u}_1 \\ \hat{u}_2 \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} \begin{pmatrix} \hat{u}_1 \\ \hat{u}_2 \end{pmatrix} \xrightarrow{\text{matrix form}} D_{uf}^2 = (\hat{u}_1 \quad \hat{u}_2) \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} \begin{pmatrix} \hat{u}_1 \\ \hat{u}_2 \end{pmatrix}$$

curvature vector

$$D_{uf}^2 = \hat{u} \cdot H_f \hat{u}$$

$$D_{uf}^2 = \hat{u}^T H_f \hat{u}$$

Hessian Matrix of func. f

$$\text{Hessian Matrix, } H = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

$$\# \text{ Rmb } \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

Properties of Hessian Matrix, H

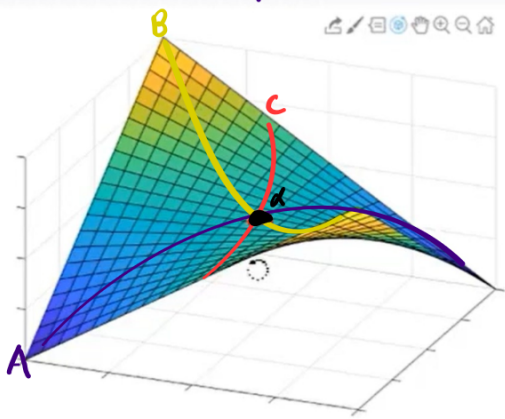
$$1^{\text{st}} \text{ column} = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} \\ \frac{\partial^2 f}{\partial x \partial y} \end{pmatrix} = \frac{\partial}{\partial x} \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \frac{\partial}{\partial x} \nabla f$$

The 1st column represents the rate of change of the gradient vector when x increases

$$2^{\text{nd}} \text{ column} = \begin{pmatrix} \frac{\partial^2 f}{\partial y \partial x} \\ \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \frac{\partial}{\partial y} \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \frac{\partial}{\partial y} \nabla f$$

The 2nd column represents the rate of change of the gradient vector when y increases

Saddle shape



In the POV of ...

- Direc. A, point a is clearly a ^[convex] max point
- Direc. B, point a is clearly a ^[concave] min point
- Direc. C, point a might be max/min, we are not sure. [Mixed curvature]

① The eigenvec. of H gives us the direc. of pure curvature, any other direction is the direc. of mixed curvature.

The corresponding e.val gives us the curvature.

- E.val +ve $\rightarrow$ As we go in e.vec direc., the shape is curving up [Min point]
- E.val -ve $\rightarrow$ As we go in e.vec direc., the shape is curving down [Max point]

[We can also use the conventional way of calc. det.]

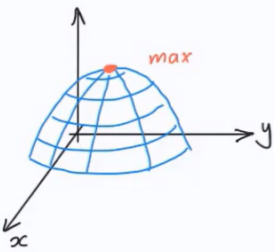
② $\det(H) = \text{Product of e.val}$ [Reason is crazy deep] [Assuming H is a 2×2 matrix w 2 e.val]

If all e.val is +ve or -ve $\rightarrow \det(H)$ is +ve $\rightarrow$ Curvature is constant [concave or convex]

If e.vals have diff signs, $\rightarrow \det(H)$ is -ve $\rightarrow$ Mixed Curvature [eg. Saddle Point]

Properties of critical points

Local maximum



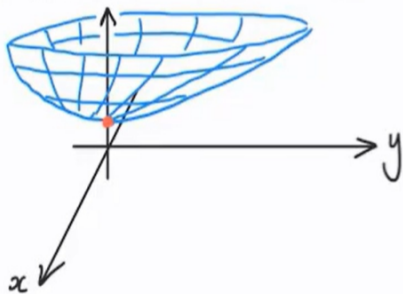
even the original axis of x & y

• $D_{\hat{u}}^2 f < 0$ for all $\hat{u}$ [Curves down in every direction]

↳ All e.val is -ve

↳ $\det(H) > 0$ [Assuming H is a 2×2 matrix, thus having 2 e.val]

Local minimum



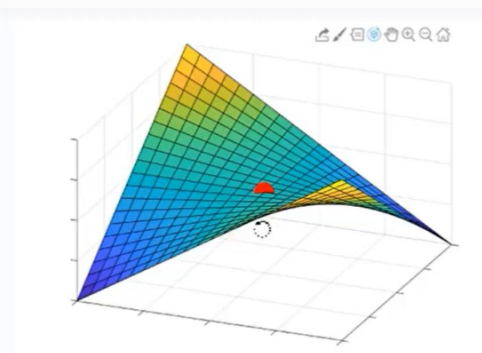
even the original axis of x & y

• $D_{\hat{u}}^2 f > 0$ for all $\hat{u}$ [Curves up in every direction]

↳ All e.val is +ve

↳ $\det(H) > 0$ [Assuming H is a 2×2 matrix, thus having 2 e.val]

Saddle



• $D_{\hat{u}}^2 f > 0$ for at least one direc., $\hat{u}$

AND

$D_{\hat{u}}^2 f < 0$ for at least one direc., $\hat{u}$

↳ E.val is a mix of +ve & -ve

↳ $\det(H) < 0$

[Assuming H is a 2×2 matrix, thus having 2 e.val]

Example 1

Local maxima/minima
OR
Saddle point

Characterise the critical points of $f(x,y) = -(x^2-1)(y^2-1)$

After setting $\nabla f = \vec{0}$, we get 5 critical points

$(0,0)$, $(-1,-1)$, $(-1,1)$, $(1,-1)$ and $(1,1)$

$$\frac{\partial f}{\partial x} = -2x(y^2-1)$$

$$\frac{\partial f}{\partial y} = -2y(x^2-1)$$

$$\frac{\partial^2 f}{\partial x^2} = -2(y^2-1)$$

$$\frac{\partial^2 f}{\partial y \partial x} = -4xy$$

$$\frac{\partial^2 f}{\partial y^2} = -2(x^2-1)$$

At $(0,0)$

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial x \partial y} = 0, \quad \frac{\partial^2 f}{\partial y^2} = 2$$

$$\det(H) = \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

$$= (2)(2) - 0$$

$$= 4 > 0$$

$\therefore$ Since $\frac{\partial^2 f}{\partial x^2}$ and $\frac{\partial^2 f}{\partial y^2} > 0$ AND

$\det(H) > 0$ meaning it is not a saddle point,

$(0,0)$ is a local minimum

This is not enough to prove local minimum. It still could be a saddle point if in one direc. $\frac{\partial^2 f}{\partial u^2} < 0$

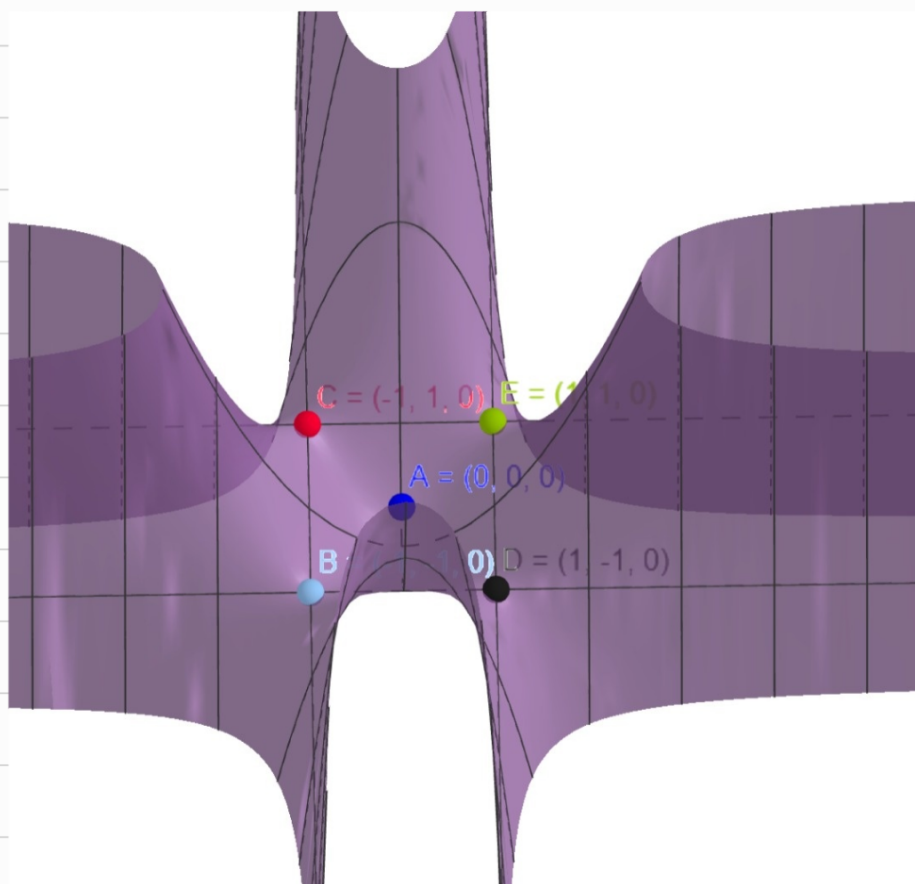
At $(\pm 1, \pm 1)$

$$\frac{\partial^2 f}{\partial x^2} = 0, \quad \frac{\partial^2 f}{\partial y \partial x} = \pm 4, \quad \frac{\partial^2 f}{\partial y^2} = 0$$

$$\det(H) = (0)(0) - (\pm 4)^2$$

$$= -16 < 0$$

$\therefore$ Since $\det(H) < 0$, $(\pm 1, \pm 1)$ are all saddle points

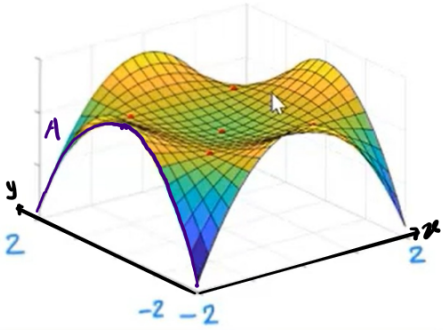


Absolute maxima and minima

- Max/Min of a function on a domain

Example 1

What is the absolute max and min of $f(x,y) = -(x^2-1)(y^2-1)$ on the domain $|x| < 2$, $|y| < 2$



From previous example, we found a local minimum at $(0,0)$

$$f(0,0) = -(0-1)(0-1) = -1$$

The saddle points is definitely not the absolute max/min becu of the mixed curvature it has

Checking the boundaries for any max/min

→ Checking curve A where

$$|y| \leq 2$$

$$x = -2, -2 \leq y \leq 2$$

$$f(-2, y) = -(4-1)(y^2-1) = -3(y^2-1)$$

$$\text{Let } g(y) = -3(y^2-1)$$

$$\frac{dg}{dy} = -6y, \text{ when } \frac{dg}{dy} = 0, y = 0$$

$$\frac{d^2g}{dy^2} = -6 < 0 \text{ (max)}$$

$$f(-2, 0) = -(4-1)(0-1) = 3$$

Checking corners:

$$f(-2, -2) = -(4-1)(4-1) = -9$$

All other corners gives -9

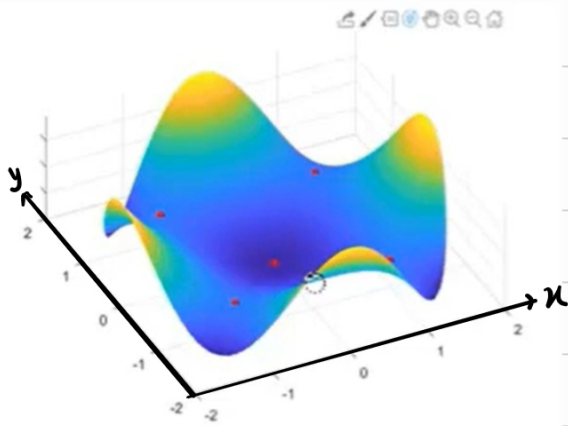
∴ Absolute maxima = 3

Absolute minima = -9

Example 2

What is the absolute max and min of $f(x,y) = -(x^2-1)(y^2-1)$ on the domain $x^2 + y^2 \leq 4$

From Example 1, local minimum at $(0,0)$: $f(0,0) = -1$



Checking the boundaries for any max/min

$$x^2 + y^2 = 4$$

$$y^2 = 4 - x^2$$

When $y^2 = 4 - x^2$

$$\begin{aligned} f(x, 4-x^2) &= -(x^2-1)(4-x^2-1) \\ &= -(x^2-1)(3-x^2) \end{aligned}$$

$$\begin{aligned} \text{Let } g(x) &= -(x^2-1)(3-x^2) \\ &= x^4 - 4x^2 + 3 \end{aligned}$$

$$\frac{dg}{dx} = 4x^3 - 8x$$

$$\frac{d^2g}{dx^2} = 12x^2 - 8$$

when $\frac{dg}{dx} = 0$

$$4x(x^2 - 2) = 0$$

$$x = 0 \quad \text{or} \quad x = \pm\sqrt{2}$$

When $x = 0$

$$\frac{d^2g}{dx^2} = -8 < 0 \quad (\text{maximum})$$

$$y^2 = 4 - 0$$

$$y = \pm 2$$

$$\therefore f(0, \pm 2) = 3$$

When $x = \pm\sqrt{2}$

$$\frac{d^2g}{dx^2} = 16 > 0 \quad (\text{minimum})$$

$$y^2 = 4 - 2$$

$$y = \pm\sqrt{2}$$

$$\therefore f(\pm\sqrt{2}, \pm\sqrt{2}) = -1$$

Checking corners:

When $x = \pm 2, y = 0$

$$f(\pm 2, 0) = 3$$

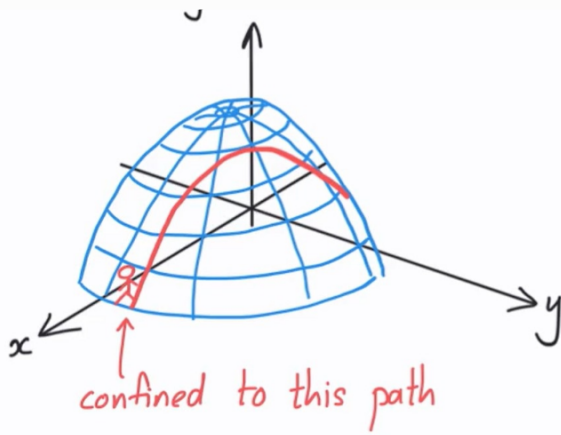
When $x = 0, y = \pm 2$

$$f(0, \pm 2) = 3$$

$\therefore$ Absolute Maxima : 3

Absolute Minima : -1

Lagrange multipliers



* Remb that on a given contour/height, $f(x,y)$ is constant.

$$\hookrightarrow \frac{df}{du} = 0$$

$$\hat{u} \cdot \nabla f = 0$$

$$\therefore \nabla f \perp \hat{u}$$

path taken

At the stationary point A, $\hat{u}$ is tangential to a contour of $f(x,y)$, having the tangential line l

[When 2 curves touch, that point is obviously the point of tangency]

$$\therefore \nabla f \perp \hat{u} \Rightarrow \nabla f \perp l$$

Notice that the path given $[g(x,y) = \text{const}]$ is a contour for the function $g(x,y)$.

At the same stationary point A, $\hat{u}$ gives the same tangential line, l .

$$\therefore \nabla g \perp \hat{u} \Rightarrow \nabla g \perp l$$

[Dirac - excluding sense]

Since $\nabla f \perp l$ and $\nabla g \perp l$ at A, it means that the span of ∇f & ∇g is the same.

$\therefore \nabla f$ is coincident on ∇g

$$\Rightarrow \boxed{\nabla f = \lambda \nabla g}$$

Lagrange multiplier.

$$2x(1+2y) = \lambda 2x \Rightarrow 2x(1+2y) - \lambda 2x = 0$$

$$2y + x^2 = \lambda(2y)$$

$$x^2 + y^2 = 1$$

$$2x[(1+2y) - \lambda] = 0$$

$$x = 0 \quad \text{or} \quad y = \frac{\lambda}{2}$$

Example 1:

Minimise $x^2 + y^2$ subject to $x + 2y = 1$ *path we want to take*

$$y = -\frac{1}{2}x + \frac{1}{2}$$

Let: $f(x, y) = x^2 + y^2$

$$g(x, y) = x + 2y$$

At a stationary point, ∇f is in the same span as ∇g

$$\therefore \nabla f = \lambda \nabla g$$

$$\begin{pmatrix} 2x \\ 2y \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} \implies 2x = \lambda \implies x = \frac{1}{2}\lambda \quad \text{--- ①}$$

$$y = \lambda \quad \text{--- ②}$$

When $g(x, y) = 1$,

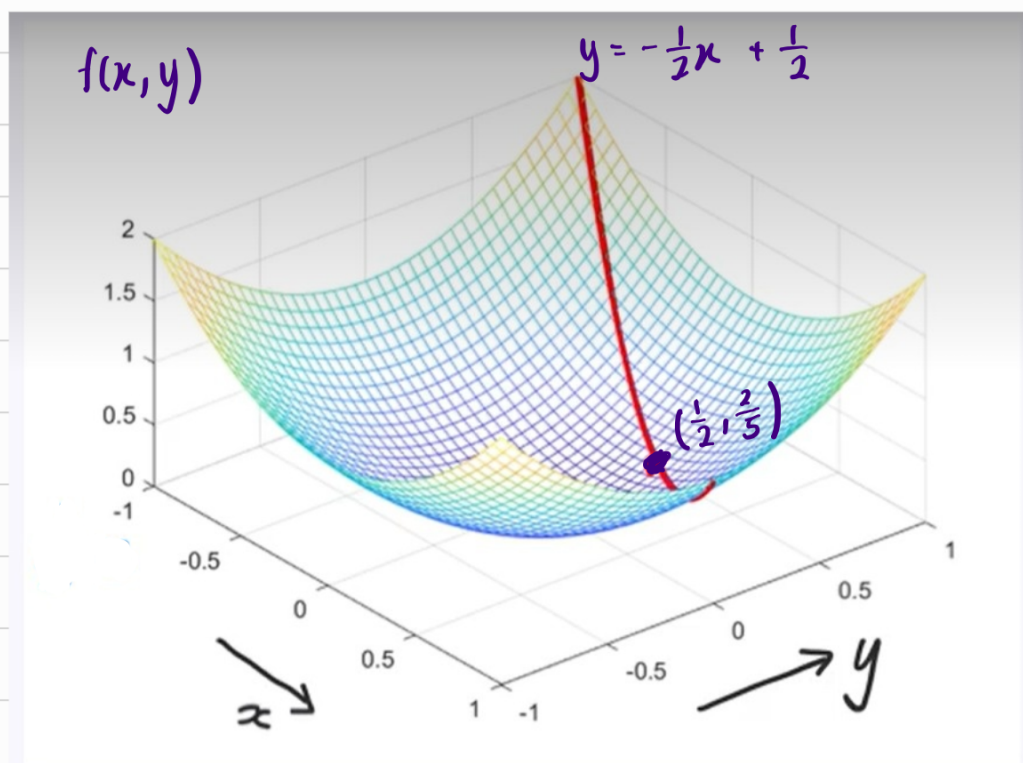
$$\frac{1}{2}\lambda + 2\lambda = 1, \text{ using ① \& ②}$$

$$\lambda = \frac{2}{5}$$

$$\therefore x = \left(\frac{1}{2}\right)\left(\frac{2}{5}\right) = \frac{1}{5} \neq$$

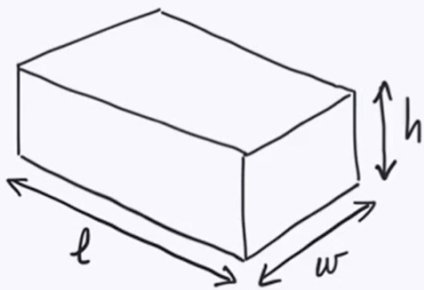
$$y = \frac{2}{5} \neq$$

$$\therefore \text{Minimum of } f(x, y) \text{ is } f\left(\frac{1}{5}, \frac{2}{5}\right) = \frac{1}{25} + \frac{4}{25} = \frac{1}{5} \neq$$



Example 2

Minimise the surface area of a cuboid subject to the constraint that the volume is 1.



Surface area

$$f(l, w, h) = 2lh + 2wh + 2wl$$

Volume

$$g(l, w, h) = lwh$$

Using the same theory from Example 1:

$$\nabla f = \lambda \nabla g$$

$$\begin{pmatrix} \partial f / \partial l \\ \partial f / \partial w \\ \partial f / \partial h \end{pmatrix} = \lambda \begin{pmatrix} \partial g / \partial l \\ \partial g / \partial w \\ \partial g / \partial h \end{pmatrix}$$

$$\begin{pmatrix} 2h + 2w \\ 2h + 2l \\ 2l + 2w \end{pmatrix} = \lambda \begin{pmatrix} wh \\ lh \\ lw \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2h + 2w = \lambda wh & \text{--- (1)} \\ 2h + 2l = \lambda lh & \text{--- (2)} \\ 2l + 2w = \lambda lw & \text{--- (3)} \end{cases}$$

$$\text{When } g(l, w, h) = 1 \Rightarrow lwh = 1$$

$$(1 \times l) : 2hl + 2wl = \lambda \quad \text{--- (4)}$$

$$(2 \times w) : 2hw + 2lw = \lambda \quad \text{--- (5)}$$

$$(3 \times h) : 2lh + 2wh = \lambda \quad \text{--- (6)}$$

Comparing (4) & (5),

$$2hl = 2hw$$

$$\therefore h = 0 \text{ or } l = w \quad \# \\ \text{(reject)} \\ \therefore lwh = 1$$

Comparing (5) & (6),

$$2lw = 2lh$$

$$l = 0 \text{ or } h = w \quad \# \\ \text{(reject)} \\ \therefore lwh = 1$$

$$\therefore l = w = h$$

$$\therefore \text{From } g(l, w, h), l(l)(l) = 1 \\ l = 1$$

$$\therefore l = 1, w = 1, h = 1$$

$$\therefore f(1, 1, 1) = 2 + 2 + 2 \\ = 6 \quad \#$$

Example 3:

Use lagrange multipliers to find the minimum distance btwn the graph of function
 $f(x, y) = x + 2y^2 + 3$
and the point $(3, 0, 0)$.

Distance btwn (x, y, z) and $(3, 0, 0)$ is

$$D(x, y, z) = \sqrt{(x-3)^2 + (y-0)^2 + (z-0)^2}$$
$$= \sqrt{(x-3)^2 + y^2 + z^2} \quad [\text{Objective function}]$$

$$D^2(x, y, z) = (x-3)^2 + y^2 + z^2$$

Constraint : (x, y, z) must satisfy $z = x + 2y^2 + 3$
 $\implies x + 2y^2 - z + 3 = 0$
 $g(x, y, z) = 0$

If (x, y, z) minimises D , (x, y, z) minimises D^2

So to find (x, y, z) , we can take ...

$$\nabla D^2 = \lambda \nabla g$$
$$\begin{pmatrix} 2(x-3) \\ 2y \\ 2z \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 4y \\ -1 \end{pmatrix}$$

$$\begin{cases} 2(x-3) = \lambda & \text{--- ①} \\ 2y = (4y)\lambda & \text{--- ②} \\ 2z = -\lambda & \text{--- ③} \end{cases}$$

Using ②,

$$2y[1 - 2\lambda] = 0$$

$$y = 0 \quad \text{or} \quad \lambda = 1/2$$

Case 1: When $y = 0$,

$$g(x, 0, z) = x - z + 3 = 0$$

$$z = x + 3$$

$$2z = 2(x + 3)$$

$$-\lambda = 2x + 6$$

$$-[2(x-3)] = 2x + 6$$

$$0 = 4x$$

$$x = 0$$

$$\therefore z = 3$$

$$\therefore D(0, 0, 3) = \sqrt{(0-3)^2 + 0^2 + 3^2}$$
$$= 3\sqrt{2} \quad \#$$

Case 2: when $\lambda = 1/2$

Using eqn ③,
 $z = -1/4$

Using eqn ①,
 $x - 3 = 1/4$
 $x = 13/4$

$$g(13/4, y, -1/4) = 13/4 + 2y^2 - (-1/4) + 3$$

$$0 = \frac{13}{2} + 2y^2$$

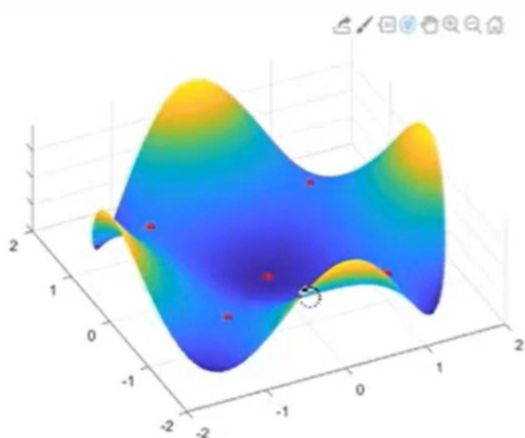
$$y^2 = -\frac{13}{4} \quad \#$$

(reject)

$\therefore$ Minimum distance is $3\sqrt{2}$ #

Example 4: [Same ques on pg 8]

What is the absolute max and min of $f(x,y) = -(x^2-1)(y^2-1)$ on the domain $x^2 + y^2 \leq 4$



Checking the boundary for max/min

$$x^2 + y^2 = 4$$

Let $g(x,y) = x^2 + y^2$

$$\nabla f = \lambda \nabla g$$

$$\begin{pmatrix} -2x(y^2-1) \\ -2y(x^2-1) \end{pmatrix} = \lambda \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow \begin{cases} -2x(y^2-1) = \lambda 2x & \text{--- ①} \\ -2y(x^2-1) = \lambda 2y & \text{--- ②} \end{cases}$$

When $g(x,y) = 4$, $x^2 + y^2 = 4$ --- ③

From ①,

$$x = 0 \quad \text{or} \quad -(y^2-1) = \lambda$$

From ②,

$$y = 0 \quad \text{or} \quad -(x^2-1) = \lambda$$

To satisfy $g(x,y) = x^2 + y^2 = 4$, x and y can't be both zero.

$$\therefore x = 0 \quad \text{and} \quad -(x^2-1) = \lambda \\ \Rightarrow \lambda = 1$$

[using ②]

$$\therefore \text{When } x = 0, \quad 0^2 + y^2 = 4 \\ y = \pm 2$$

$$f(0, \pm 2) = 3$$

(OR)

$$y = 0 \quad \text{and} \quad -(y^2-1) = \lambda \\ \Rightarrow \lambda = 1$$

[using ③]

$$\therefore \text{When } y = 0, \quad x^2 + 0^2 = 4 \\ x = \pm 2$$

$$f(\pm 2, 0) = 3$$

(OR)

$$-(y^2-1) = \lambda \quad \text{and} \quad -(x^2-1) = \lambda \\ \Rightarrow y^2 = 1 - \lambda \quad \text{and} \quad x^2 = 1 - \lambda$$

$$\therefore (1-\lambda) + (1-\lambda) = 4 \quad \text{[using ③]} \\ \lambda = -1$$

$$\therefore y^2 = 2$$

$$y = \pm \sqrt{2}$$

$$x^2 = 2$$

$$x = \pm \sqrt{2}$$

$$f(\pm \sqrt{2}, \pm \sqrt{2}) = -1$$

$\therefore$ Absolute max is 3
Absolute min is -1

Parametric functions

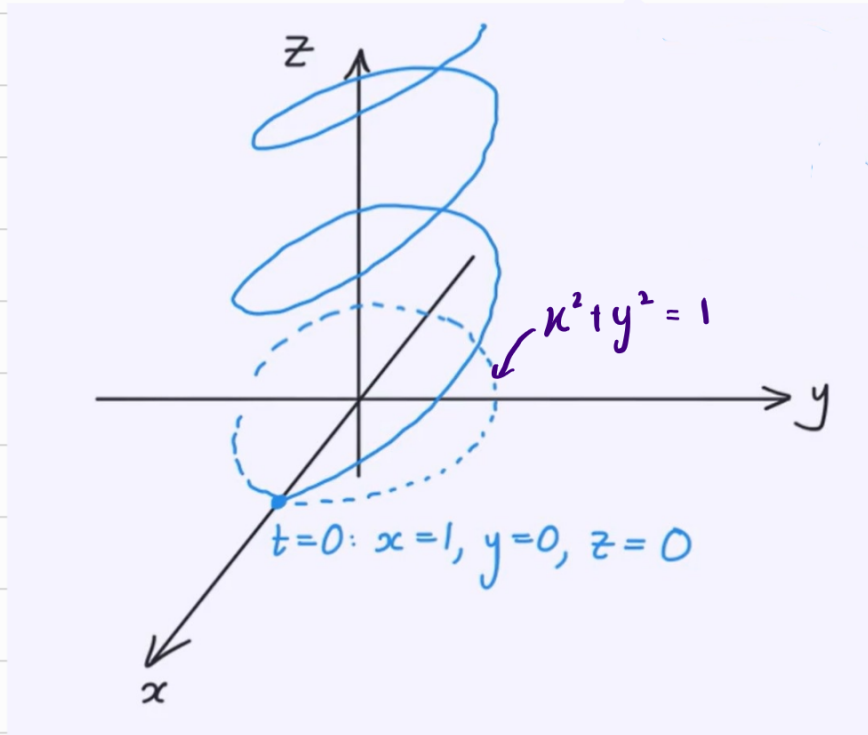
Example 1:

What is the curve $x(t) = \cos(t)$, $y(t) = \sin(t)$, $z(t) = t$ for $0 \leq t \leq \infty$

From the identity $\cos^2(t) + \sin^2(t) = 1$

$$x^2 + y^2 = 1$$

So x is related to y by a circle at centre $(0,0)$ w radius 1



Example 2:

1 parameter

Find a parameterisation for the curve given by the intersection of the plane $x+y+z=0$ and the parabolic cylinder $z=y^2$

$$\begin{aligned} \text{Set } y=t, \text{ then } z=t^2 \\ \left. \begin{aligned} x &= -y-z \\ &= -t-t^2 \end{aligned} \right\} \begin{aligned} x(t) &= -t-t^2 \\ y(t) &= t \\ z(t) &= t^2 \end{aligned} \end{aligned}$$

Example 3:

1 parameter

Find a parameterisation for the curve $x=2$, $y^2+2y+z^2=3$

Rewriting the equation : $y^2+2y+z^2=3$

$$(y^2+2y+1)+z^2=4$$

$$(y+1)^2+z^2=4$$

$$\frac{(y+1)^2}{4} + \frac{z^2}{4} = 1 \quad \text{--- ①}$$

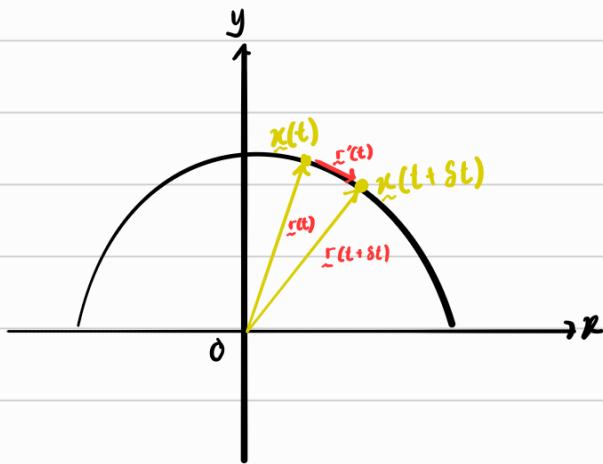
Comparing ① w $\cos^2(t) + \sin^2(t) = 1$,

$$\Rightarrow \cos(t) = \frac{y+1}{2}, \quad \sin(t) = \frac{z}{2}$$

$$\Rightarrow y(t) = -1 + 2\cos(t), \quad z(t) = 2\sin(t)$$

$$\therefore x(t) = 2, \quad y(t) = -1 + 2\cos(t), \quad z(t) = 2\sin(t), \quad \text{for } 0 \leq t < 2\pi$$

Tangent Vector



$$\text{tangent vector, } \underline{r}'(t) = \lim_{\delta t \rightarrow 0} \frac{\underline{x}(t+\delta t) - \underline{x}(t)}{\delta t}$$

$$\underline{r}'(t) = \frac{d\underline{x}}{dt}$$

* $\frac{dy}{dx}$ finds the (gradient of tangent line) instantaneous gradient by taking the gradient btwn 2 very close points

$\frac{dy}{dx}$ finds the direc. of the tangent line by taking the gradient btwn 2 very close position vec.

Example 1:

What is the tangent vector of $\underline{x}(t) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + t \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$?

$$\underline{x}(t) = \begin{pmatrix} 1 + 4t \\ 2 + 5t \\ 3 + 6t \end{pmatrix}$$

$$\text{tangent vector} = \frac{d\underline{x}}{dt} = \frac{d}{dt} \begin{pmatrix} 1 + 4t \\ 2 + 5t \\ 3 + 6t \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \quad \#$$

Parametrising Surfaces

- Surfaces needs 2 parameters to describe all the points on them

2 Parameters

Consider the vector plane eqn

$$\underline{r} = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} + r \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 + r - 2s \\ 0 + r + 0 \\ 0 + 0 + s \end{pmatrix}$$

$$(x, y, z) = (5 + r - 2s, r, s)$$

$$x(r, s) = 5 + r - 2s$$

$$y(r, s) = r$$

$$z(r, s) = s$$

Example 1:

Consider the cylinder $\underline{x}(\theta, t) = \begin{pmatrix} 3 \cos \theta \\ 3 \sin \theta \\ t \end{pmatrix}$

$$x(\theta, t) = 3 \cos \theta$$

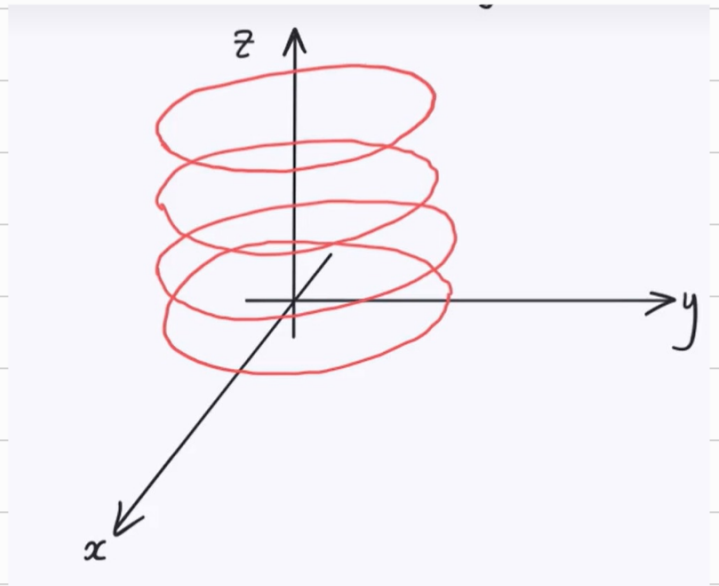
$$y(\theta, t) = 3 \sin \theta$$

From the identity, $\cos^2 \theta + \sin^2 \theta = 1$

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$

$$x^2 + y^2 = 3^2$$

$\therefore$ Circle w radius 3



Therefore, for every $z(t) = t$, we draw a circle.

This will make a cylinder.

Example 2:

What is the surface given by

$$x = \cos(s) \cos(t) \quad y = \cos(s) \sin(t) \quad z = \sin(s) \quad \text{for } -\pi \leq s \leq \pi$$
$$0 \leq t < 2\pi$$

$$x^2 + y^2 = \cos^2(s) \cos^2(t) + \cos^2(s) \sin^2(t)$$

$$= \cos^2(s) [\cos^2(t) + \sin^2(t)]$$

$$x^2 + y^2 = \cos^2(s)$$

$$\text{Now, } x^2 + y^2 + z^2 = \cos^2(s) + \sin^2(s)$$

$$x^2 + y^2 + z^2 = 1$$

$\therefore$ Sphere w radius 1 centered at $(0, 0, 0)$

Example 3:

2 parameters

Find a parameterisation for the surface given by $x + y = z^3$

$$\left. \begin{array}{l} \text{Let } y = s \\ z = t \\ \therefore x = t^3 - s \end{array} \right\} \begin{array}{l} \therefore x(s, t) = t^3 - s \\ y(s, t) = s \\ z(s, t) = t \end{array} \quad \text{for } t \in \mathbb{R}, s \in \mathbb{R}$$

Example 4:

Find a parameterisation for $x^2 + \frac{1}{4}y^2 + \frac{1}{9}z^2 = 1$

$$\begin{aligned} \text{Let } x^2 + \frac{1}{4}y^2 + \frac{1}{9}z^2 &= 1 \\ (x^2 + \frac{1}{4}y^2) + \frac{1}{9}z^2 &= \cos^2(s) + \sin^2(s) \end{aligned}$$

By comparison,

$$\begin{aligned} x^2 + \frac{1}{4}y^2 &= \cos^2(s) \\ &= \cos^2(s) [\cos^2(t) + \sin^2(t)] \\ &= \cos^2(s) \cos^2(t) + \cos^2(s) \sin^2(t) \end{aligned}$$

$$\begin{aligned} \frac{1}{9}z^2 &= \sin^2(s) \\ z &= 3 \sin(s) \end{aligned}$$

By comparison,

$$\begin{aligned} x &= \cos(s) \cos(t) \\ y &= 2 \cos(s) \sin(t) \end{aligned}$$

$$\therefore \left. \begin{array}{l} x(s, t) = \cos(s) \cos(t) \\ y(s, t) = 2 \cos(s) \sin(t) \\ z(s, t) = 3 \sin(s) \end{array} \right\} \begin{array}{l} \text{for } s \in [-\pi, \pi] \\ t \in [0, 2\pi) \end{array}$$